

EPINAL-DOGNEVILLE AP, France

WMO: 071730

Lat: 48.2107N

Lon: 6.4517E

Elev: 320

StdP: 97.54

Time Zone: 1.00 (EUC)

Period: 90-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.0	-7.7	-11.9	1.4	-8.9	-9.9	1.7	-6.3	11.2	9.1	10.1	8.3	0.7	110	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	30.4	19.5	28.6	19.2	26.9	18.7	21.0	28.0	20.2	26.6	19.4	25.3	3.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	14.0	23.4	17.9	13.4	22.5	17.2	12.8	21.7	61.9	28.1	59.1	26.6	56.5	25.3	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.5	6.6	DB	-13.6	33.0	3.0	2.0	-15.7	34.4	-17.4	35.5	-19.1	36.6	-21.2	38.1	(4)
(5)			WB	-13.4	22.6	3.2	0.6	-15.7	23.0	-17.6	23.4	-19.4	23.7	-21.7	24.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	1.4	2.6	5.8	9.1	13.1	16.9	18.6	18.3	14.8	10.8	5.4	1.6
	DBStd	7.25	4.67	4.52	4.24	3.41	3.43	3.67	2.97	2.95	3.18	3.77	3.70	4.13
	HDD10.0	1124	267	208	139	57	11	1	0	0	2	37	142	260
	HDD18.3	3190	524	441	388	277	166	71	33	38	114	233	387	518
	CDD10.0	1091	1	1	9	29	107	208	267	256	145	63	5	1
	CDD18.3	117	0	0	0	0	3	28	43	36	7	0	0	0
	CDH23.3	1437	0	0	0	9	56	317	520	452	81	2	0	0
	CDH26.7	406	0	0	0	0	8	85	152	153	7	0	0	0
Wind	WSAvg	2.6	3.0	3.0	2.9	2.5	2.7	2.4	2.5	2.3	2.1	2.4	2.6	3.0
Precipitation	PrecAvg	1130	101	90	81	76	96	86	87	88	90	107	112	116
	PrecMax	1512	231	171	235	190	161	164	187	188	166	209	274	267
	PrecMin	792	25	21	10	16	21	27	26	9	24	17	23	16
	PrecStd	162	50	45	55	44	33	36	38	43	45	52	61	62
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	15.6	19.5	24.6	27.8	31.8	32.7	35.5	27.9	23.1	16.7	11.3
		MCWB	10.1	9.1	11.7	15.0	18.9	20.1	20.1	19.8	18.3	16.7	12.5	9.6
	2%	DB	10.9	12.9	17.5	21.8	24.8	29.0	30.1	30.7	25.7	20.9	14.3	10.1
		MCWB	9.1	8.4	10.8	13.7	16.1	19.6	19.5	19.2	18.2	15.7	11.5	9.1
	5%	DB	9.4	11.2	15.1	19.3	22.9	27.1	28.5	28.1	23.7	18.9	12.6	9.0
		MCWB	7.7	7.8	9.7	12.1	15.2	19.4	19.3	18.8	17.1	15.0	10.7	8.0
	10%	DB	8.2	9.6	13.0	17.0	20.9	24.9	26.6	25.9	21.5	17.0	11.0	7.6
		MCWB	6.6	7.3	9.2	11.2	14.4	18.5	18.7	18.3	16.4	14.2	9.6	6.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	10.1	13.0	16.3	20.0	21.8	21.8	21.6	20.3	17.7	13.5	10.1
		MCDB	11.7	13.2	17.9	23.1	25.2	26.7	28.9	30.2	25.7	21.5	15.1	10.8
	2%	WB	9.4	9.2	11.6	14.2	18.2	20.7	21.0	20.5	19.1	16.3	11.9	9.2
		MCDB	10.7	11.7	15.4	20.3	23.6	26.8	28.6	27.6	23.9	19.5	13.9	10.1
	5%	WB	7.9	8.3	10.6	12.8	16.2	19.8	20.1	19.6	18.0	15.3	10.8	8.1
		MCDB	9.2	10.8	14.1	17.9	20.4	25.8	26.7	26.1	22.2	18.2	12.5	8.8
	10%	WB	6.8	7.3	9.5	11.5	15.1	18.9	19.1	18.7	16.9	14.3	9.6	6.8
		MCDB	8.1	9.4	12.7	16.0	19.2	24.3	24.9	24.3	20.5	16.6	10.9	7.6
Mean Daily Temperature Range	5% DB	MDBR	6.1	7.7	9.7	11.4	11.1	11.1	11.2	11.3	11.1	9.1	6.4	5.2
		MCDBR	5.1	11.9	14.6	17.1	15.5	15.5	15.8	17.2	14.8	12.2	9.2	4.9
		MCWBR	4.0	7.8	8.9	9.5	7.8	7.0	6.7	7.2	7.8	7.6	6.5	4.1
	5% WB	MCDBR	4.6	9.0	12.2	15.1	12.9	13.5	13.7	13.8	12.3	10.7	7.8	4.1
		MCWBR	4.0	6.4	7.7	8.6	7.2	6.5	6.2	6.6	7.0	7.0	5.8	3.5
Clear-Sky Solar Irradiance	taub	0.309	0.332	0.359	0.391	0.400	0.406	0.402	0.398	0.373	0.357	0.326	0.306	
	taud	2.445	2.389	2.332	2.261	2.278	2.294	2.300	2.329	2.375	2.410	2.446	2.437	
	Ebn at Noon	764	822	853	857	863	857	855	842	828	780	733	717	
	Edh at Noon	65	85	105	125	128	127	124	116	100	83	65	58	
All-Sky Solar Radiation	RadAvg	0.97	1.73	2.96	4.31	5.07	5.88	5.51	4.79	3.65	2.20	1.12	0.78	
	RadStd	0.14	0.30	0.46	0.61	0.55	0.51	0.53	0.48	0.36	0.18	0.12	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	+0.32	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page